II. Riemann hypothesis.

1. Some review & preview

Def II 1.1 Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ $(s>1)$ has an analytic continuation over the whole complex plane.

Thm II 1.1 (i) $\zeta(s)$ is a meromorphic function with a pole at $s=1$. with residue 1.

(ii) $\zeta(s)$ satisfies $\pi^{-\frac{1}{2}s}\Gamma(\frac{1}{2}s)\zeta(s) = \pi^{-\frac{1}{2}(1-s)}\Gamma(\frac{1}{2}(1-s))\zeta(1-s)$

Define theta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{1}{2}s}\Gamma(s)\zeta(s)$. satisfies $\xi(s) = \xi(1-s)$

(iii) Let $s=\sigma+it$., $\zeta(s)$ has trivial zeros at $s=-2,-4,-6,\cdots$

Def II.1.3. (i) The region $0 \le \sigma \le 1$ is called the critical ~~line~~ strip

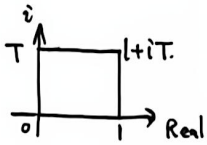
(ii) The line $\sigma=\frac{1}{2}$ is called the critical line.

Thm II.1.4 (i) $\xi(s)$ satisfies $\xi(s) = e^{A+Bs}\prod_{\rho \text{ nontrivial zeros}} (1-\frac{s}{\rho})e^{\frac{s}{\rho}}$, A, B are constants.

(ii) $\zeta(s)$ has a zero-free region described by $\sigma \ge 1 - \frac{c(\alpha)}{(\log t)^\alpha}$ for $\alpha > \frac{2}{3}$.

where $c(\alpha)$ is constant depending on α (Vinogradov).

(iii) Conjecture: the number of zeros $N(T)$ of $\zeta(s)$ with $0<t\le T$ satisfies $N(T) = \frac{T}{2\pi}\log(\frac{T}{2\pi}) - \frac{T}{2\pi} + O(\log T)$. (Proved in 1905).



(iv). Conjecture: $\psi_0(x) = \begin{cases} \psi(x) - \frac{1}{2}\Lambda(x) & \text{if } x = p^l. \\ \psi(x) & \text{otherwise.} \end{cases}$

that $\psi_0(x) = x - \sum_{\rho \text{ nontrivial zeros}} \frac{x^\rho}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \frac{1}{2}\log(1-x^{-2})$ for large x.

2 Prime number theorem

Thm II 2.1 ① $\zeta(s)$ can be continued analytically on the whole plane with a simple pole $|S=1$ with residue 1

② $\pi^{-\frac{1}{2}s}\Gamma(\frac{1}{2}s)\zeta(s) = \pi^{\frac{1}{2}(1-s)}\Gamma(\frac{1}{2}(1-s))\zeta(1-s)$ where $\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}dt$

Lemma II 2.2

(Poisson summation formula): ① $f(x)$ real, continuous, monotonic, then $\sum_{n=A}^{B}{}' f(n) = \sum_{-\infty}^{\infty}\int_A^B f(x)e^{2\pi i v x}dx$

where $\sum'$ means $f(A)\to\frac{f(A)}{2}$, $f(B)\to\frac{f(B)}{2}$

② $f\in L^1(\mathbb{R})$. $\hat{f}(y) = \int_{\mathbb{R}} f(x)e^{-2\pi i xy}dx$. If $f\in L^1(\mathbb{R})$ and both f & $\hat{f}$ have bounded variation, then $\sum_{n=-\infty}^{\infty} f(n) = \sum_{m=-\infty}^{\infty}\hat{f}(m)$.

prove ②: Let $F(x) = \sum_{m=-\infty}^{\infty} f(x+m)$., $F(x)$ has an absolutly convergent Fourier series where

$F(x) = \sum_{n=-\infty}^{\infty} C_F(n)e^{2\pi i n x}$ $\quad C_F(n) = \int_0^1 F(t)e^{-2\pi i n t}dt = \int_{-\infty}^{\infty} f(t)e^{-2\pi i n t}dt = \hat{f}(n)$ □

(Take $x=0$).

~~Back to the proof~~

Def II.2.3. $\theta(x) = \sum_{n=-\infty}^{\infty} e^{-n^2\pi x}$

Prop. II.2.4. $\sum_{n=-\infty}^{\infty} e^{-\frac{(n+\alpha)^2\pi}{x}} = x^{\frac{1}{2}}\sum_{n=-\infty}^{\infty} e^{-n^2\pi x+2\pi i n\alpha}$. $\left(\forall x>0,\ \alpha\in\mathbb{R}\right)$.

pf: By Poisson summation formula: $f(n) = e^{-\frac{(n+\alpha)^2\pi}{x}}$, $A=-N$, $B=N$.

$$\Rightarrow \lim_{N\to\infty}{\sum_{n=-N}^{N}}' e^{-\frac{(n+\alpha)^2\pi}{x}} = \lim_{N\to\infty}\sum_{v=-\infty}^{\infty}\int_{-N}^{N} e^{-\frac{(t+\alpha)^2\pi}{x}+2\pi i v t}dt \overset{e^{-(\cdot)}\text{ is rapidly decreased}}{=\!=\!=} \sum_{v=-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{(t+\alpha)^2\pi}{x}+2\pi i v t}dt$$

Let $u=\frac{t+\alpha}{x}$. Then LHS $= \sum_{v=-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-xu^2\pi+2\pi i v(xu-\alpha)}\cdot x\,du$

$= x\sum_{v=-\infty}^{\infty} e^{-2\pi i v\alpha}\int_{-\infty}^{\infty} e^{-xu^2\pi+2\pi i xuv}du$ Gauss formula.

$= x\sum_{v=-\infty}^{\infty} e^{-2\pi i v\alpha}\int_{-\infty}^{\infty} e^{-\pi x(u-iv)^2-\pi x v^2}du$ $\left(\int_{-\infty}^{\infty} e^{-\pi x(u+\beta)^2}du = Ax^{-\frac{1}{2}}\right)$.

$= x\sum_{v=-\infty}^{\infty} e^{-2\pi i v\alpha-\pi x v^2}x^{-\frac{1}{2}} = x^{\frac{1}{2}}\sum_{n=-\infty}^{\infty} e^{-n^2\pi x+2\pi i n\alpha}$. □.

Rmk. If $\alpha=0$. Then $\sum_{n=-\infty}^{\infty} e^{-\frac{n^2\pi}{x}} = x^{\frac{1}{2}}\sum_{n=-\infty}^{\infty} e^{-n^2\pi x}$ $(\forall x>0)$. $\Rightarrow$ Thus $\theta(x^{-1}) = x^{\frac{1}{2}}\theta(x)$, $\forall x>0$

For $\omega(x) = \sum_{n=1}^{\infty} e^{-n^2\pi x}$ we have $2\omega(x)+1 = \theta(x)$. $\Rightarrow$ $\omega(x^{-1}) = \frac{1}{2}+\frac{1}{2}x^{\frac{1}{2}}+x^{\frac{1}{2}}\omega(x)$

Back to the proof:

① ② $\Gamma(\frac{1}{2}s) = \int_0^\infty e^{-t}t^{\frac{1}{2}s-1}dt$ $\left(\mathrm{Re}(s)>0\right)$ $\overset{t=n^2\pi x}{=\!=\!=}$ $\int_0^\infty e^{-n^2\pi x}(n^2\pi x)^{\frac{1}{2}s-1}n^2\pi\,dx$

$= (n^2\pi)^{\frac{1}{2}s}\int_0^\infty e^{-n^2\pi x}x^{\frac{1}{2}s-1}dx$.

Thus $\pi^{-\frac{1}{2}s}\Gamma(\frac{1}{2}s) = n^s\int^\infty e^{-n^2\pi x}\cdot x^{\frac{1}{2}s-1}dx$ $\Rightarrow$ $\pi^{-\frac{1}{2}s}\Gamma(\frac{1}{2}s)n^{-s} = \int^\infty e^{-n^2\pi x}\cdot x^{\frac{1}{2}s-1}dx$

Take the summation of n from 1 to ∞, then $\pi^{-\frac{1}{2}s}\Gamma(\frac{1}{2}s)\zeta(s) = \sum_{n=1}^{\infty}\int_0^\infty e^{-n^2\pi x}\, x^{\frac{1}{2}s-1}\,dx$

$= \int_0^\infty \left(\sum_{n=1}^{\infty} e^{-n^2\pi x}\, x^{\frac{1}{2}s-1}\right)dx = \int_0^\infty x^{\frac{1}{2}s-1}\omega(x)\,dx = \int_0^1 x^{\frac{1}{2}s-1}\omega(x)\,dx + \int_1^\infty x^{\frac{1}{2}s-1}\omega(x)\,dx$

$\overset{x=\frac{1}{y}}{=} \int_1^\infty y^{1-\frac{1}{2}s}\omega(y^{-1})\cdot(-\frac{1}{y^2})\,dy + \int_1^\infty x^{\frac{1}{2}s-1}\omega(x)\,dx$

By using the previous formula: $RHS = \int_1^\infty \omega(x)\left(x^{\frac{1}{2}s-1} - x^{-\frac{1}{2}s-\frac{1}{2}}\right)dx = \pi^{\frac{1}{2}(1-s)}\Gamma\left(\frac{1}{2}(1-s)\right)\zeta(1-s)$

$-\frac{1}{s}+\frac{1}{s-1}$

)